Confirmatory Structure Analysis Demonstration

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```
library(lavaan)
library(lavaanPlot)
library(jmv)
library(psych)
```

Load in Data

```
wiscsem <- read.csv("wiscsem.csv")[,-c(1:2,13)]
cogability <- read.csv("cogability.csv")[,c(2:10)]
```

Descriptive Statistics

```
describe(wiscsem)
```

```
##          vars   n  mean   sd median trimmed  mad min max range  skew kurtosis
## info        1 175  9.50 2.91     10    9.50 2.97   3  19    16  0.08    -0.08
## comp        2 175 10.00 2.97     10    9.95 2.97   0  18    18  0.09     0.33
## arith       3 175  9.00 2.31      9    8.89 2.97   4  16    12  0.39    -0.18
## simil       4 175 10.61 3.18     11   10.62 2.97   2  18    16  0.02    -0.23
## vocab       5 175 10.70 2.93     10   10.61 2.97   2  19    17  0.27     0.29
## digit       6 175  8.73 2.70      8    8.65 1.48   0  16    16  0.27     0.07
## pictcomp    7 175 10.68 2.93     11   10.70 2.97   2  19    17 -0.07     0.29
## parang      8 175 10.37 2.66     10   10.43 2.97   2  17    15 -0.20    -0.06
## block       9 175 10.31 2.71     10   10.36 2.97   2  18    16 -0.22     0.50
## object     10 175 10.90 2.84     11   10.94 2.97   3  19    16 -0.12     0.15
##            se
## info     0.22
## comp     0.22
## arith    0.17
## simil    0.24
## vocab    0.22
## digit    0.20
## pictcomp 0.22
## parang   0.20
## block    0.20
## object   0.21
```

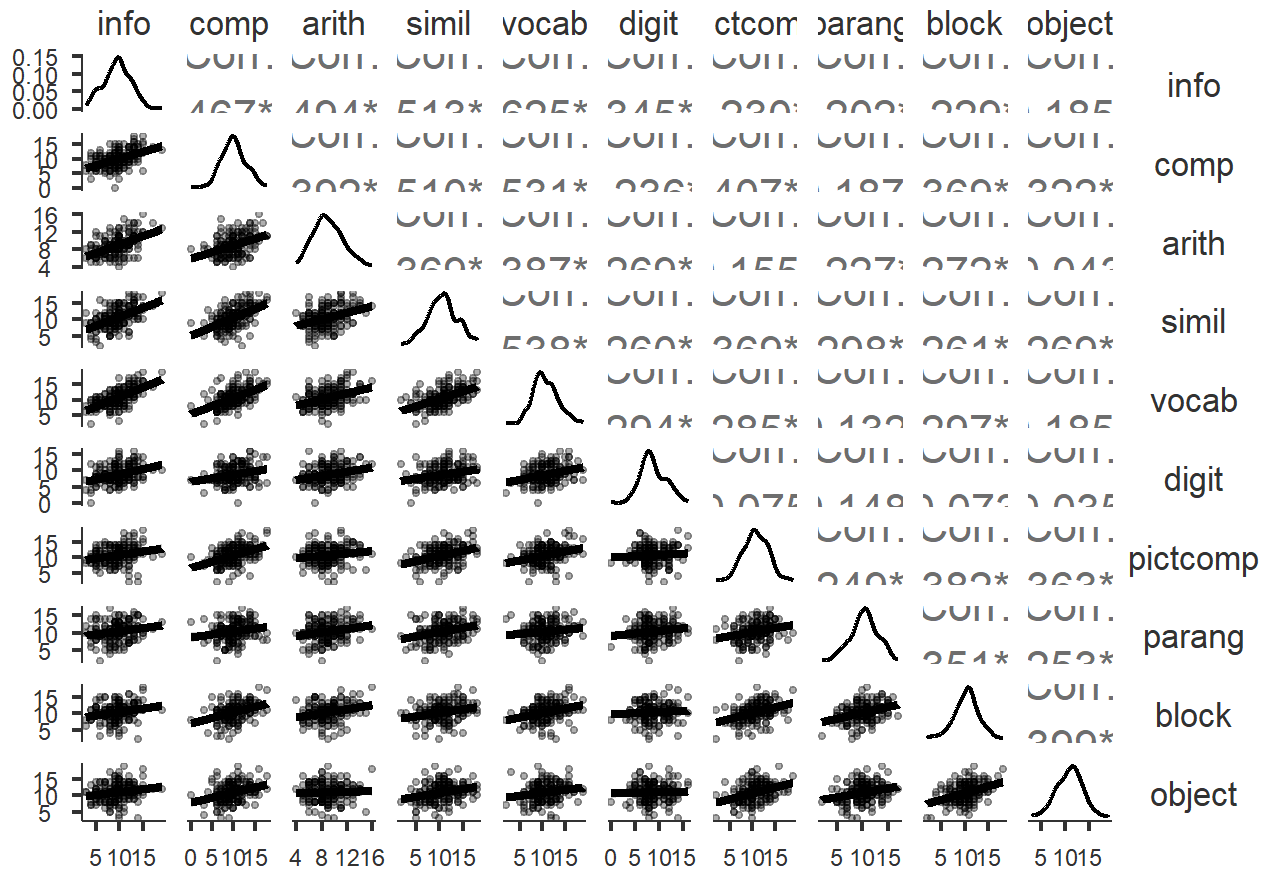
```
describe(cogability)
```

```
##    vars   n   mean    sd median trimmed   mad   min    max  range  skew
## V1    1 250  99.91 18.60 100.91  100.63 17.99 39.86 165.46 125.60 -0.29
## V2    2 250 100.21 19.89  99.44  100.04 19.43 44.66 159.19 114.54  0.06
## V3    3 250  99.21 19.36  98.88   99.26 20.52 47.98 158.36 110.38  0.01
## V4    4 250 100.51 21.55  99.66   99.99 21.13 45.82 157.00 111.17  0.20
## V5    5 250  99.99 21.19  99.80  100.26 20.79 48.95 153.97 105.02 -0.09
## V6    6 250 101.84 19.95 101.95  101.92 19.28 38.42 150.38 111.96 -0.10
## V7    7 250 101.16 20.58 101.38  100.72 19.44 44.60 167.07 122.47  0.19
## V8    8 250 102.37 20.81 103.58  102.80 21.92 46.30 145.86  99.56 -0.19
## V9    9 250 101.34 19.63 101.58  101.41 20.93 48.87 158.29 109.42 -0.03
##    kurtosis   se
## V1     0.45 1.18
## V2     0.09 1.26
## V3    -0.23 1.22
## V4    -0.35 1.36
## V5    -0.57 1.34
## V6     0.02 1.26
## V7     0.26 1.30
## V8    -0.51 1.32
## V9    -0.26 1.24
```

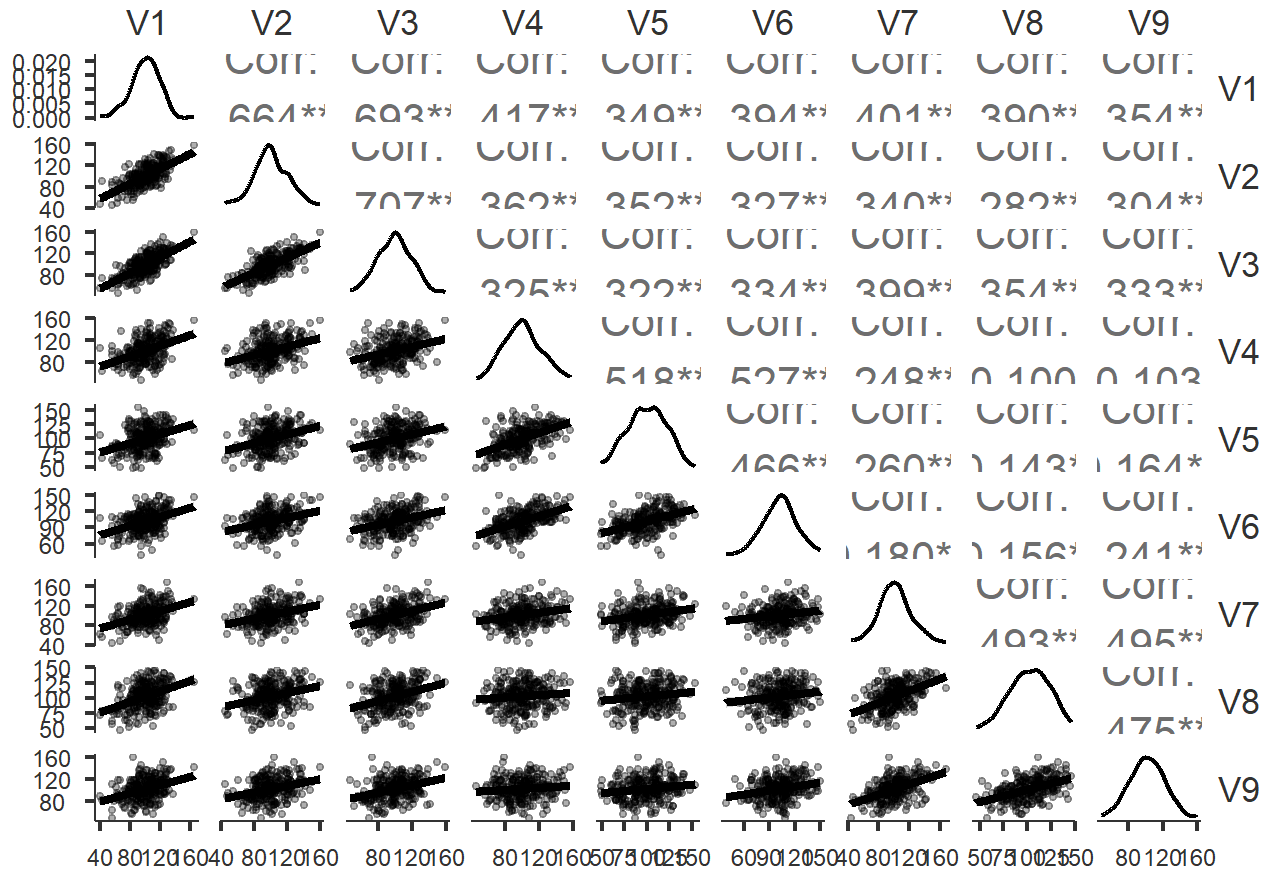
`CorrMatrix()` output

```
corrMatrix(wiscsem,plots = TRUE,plotDens = TRUE,plotStats = TRUE)$plot
```

```
## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
## ℹ Please use the `linewidth` argument instead.
## ℹ The deprecated feature was likely used in the jmvcore package.
##   Please report the issue at <https://github.com/jamovi/jmvcore/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```
corrMatrix(cogability,plots = TRUE,plotDens = TRUE,plotStats = TRUE)$plot
```



Confirmatory Factor Analysis (CFA)

Model Specification (i.e., Measurement Model)

One-factor Model:

```
onefactor <- ' # Latent variable
g =~ info + comp + arith + simil + vocab + digit + pictcomp + parang + block + object

# Residuals for manifest variables
info~~info
comp~~comp
arith~~arith
simil~~simil
vocab~~vocab
digit~~digit
pictcomp~~pictcomp
parang~~parang
block~~block
object~~object

# Variances for latent variables
g~~1*g
'
```

Two-factor (Uncorrelated) Model:

```
twofactor <- ' # Latent variables
verbal =~ info + comp + arith + simil + vocab + digit
performance =~ pictcomp + parang + block + object

# Residuals for manifest variables
info~~info
comp~~comp
arith~~arith
simil~~simil
vocab~~vocab
digit~~digit
pictcomp~~pictcomp
parang~~parang
block~~block
object~~object

# Variances for latent variables
verbal~~1*verbal
performance~~1*performance
'
```

Two-factor (Correlated) Model:

```
ctwofactor <- ' # Latent variables
verbal =~ info + comp + arith + simil + vocab + digit
performance =~ pictcomp + parang + block + object

# Latent variable correlation
verbal ~~ performance

# Residuals for manifest variables
info~~info
comp~~comp
arith~~arith
simil~~simil
vocab~~vocab
digit~~digit
pictcomp~~pictcomp
parang~~parang
block~~block
object~~object

# Variances for latent variables
verbal~~1*verbal
performance~~1*performance
'
```

Run the model, summarize the parameter estimates, and plot:

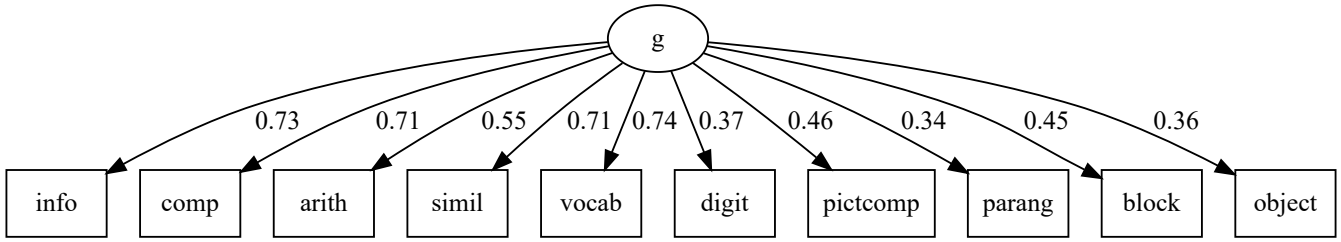
```
# Run the model: One Factor Model
fit <- lavaan(onefactor,
              data = wiscsem,
              std.lv = TRUE)

# Get parameter estiamtes and fit statistics
summary(fit,
        fit.measures = TRUE,
        standardized = TRUE)
```

```
## lavaan 0.6-19 ended normally after 23 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        20
##
##   Number of observations                           175
##
## Model Test User Model:
##
##   Test statistic                               101.635
##   Degrees of freedom                                35
##   P-value (Chi-square)                           0.000
##
## Model Test Baseline Model:
##
##   Test statistic                               503.222
##   Degrees of freedom                                45
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.855
##   Tucker-Lewis Index (TLI)                       0.813
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -4082.864
##   Loglikelihood unrestricted model (H1)      -4032.046
##
##   Akaike (AIC)                                8205.727
##   Bayesian (BIC)                              8269.023
##   Sample-size adjusted Bayesian (SABIC)       8205.689
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.104
##   90 Percent confidence interval - lower         0.081
##   90 Percent confidence interval - upper         0.128
##   P-value H_0: RMSEA <= 0.050                    0.000
##   P-value H_0: RMSEA >= 0.080                    0.957
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.081
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
```

```
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   g =~
##     info              2.110    0.203   10.417    0.000    2.110    0.727
##     comp              2.102    0.208   10.116    0.000    2.102    0.711
##     arith             1.271    0.172    7.370    0.000    1.271    0.552
##     simil             2.265    0.223   10.168    0.000    2.265    0.714
##     vocab             2.172    0.202   10.734    0.000    2.172    0.743
##     digit             1.004    0.212    4.726    0.000    1.004    0.372
##     pictcomp          1.355    0.225    6.011    0.000    1.355    0.463
##     parang            0.898    0.210    4.272    0.000    0.898    0.339
##     block             1.229    0.209    5.890    0.000    1.229    0.455
##     object            1.017    0.224    4.542    0.000    1.017    0.359
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .info              3.981    0.529    7.523    0.000    3.981    0.472
##    .comp              4.325    0.563    7.683    0.000    4.325    0.495
##    .arith             3.677    0.425    8.643    0.000    3.677    0.695
##    .simil             4.946    0.646    7.656    0.000    4.946    0.491
##    .vocab             3.833    0.522    7.337    0.000    3.833    0.448
##    .digit             6.263    0.689    9.095    0.000    6.263    0.861
##    .pictcomp          6.725    0.754    8.913    0.000    6.725    0.786
##    .parang            6.226    0.681    9.145    0.000    6.226    0.885
##    .block             5.791    0.648    8.933    0.000    5.791    0.793
##    .object            7.007    0.769    9.116    0.000    7.007    0.871
##     g                 1.000                               1.000    1.000
```

```
lavaanPlot(fit,
           coefs = TRUE,
           stand = TRUE)
```



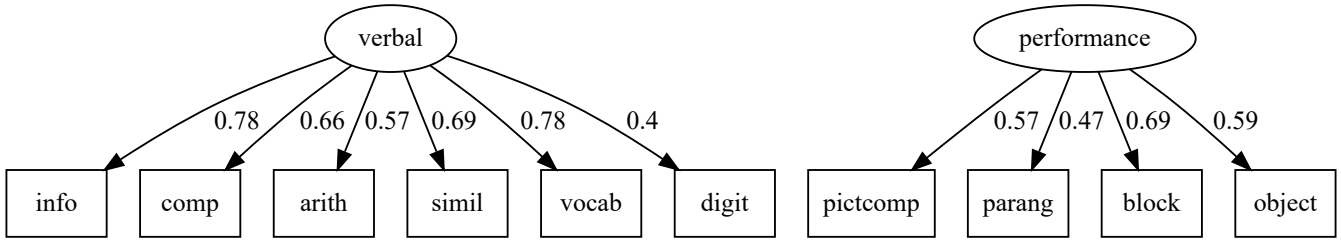

```
# Run the model: Two Factor Model
fit2 <- lavaan(twofactor,
               data = wiscsem,
               std.lv = TRUE)

# Get parameter estiamtes and fit statistics
summary(fit2,
        fit.measures = TRUE,
        standardized = TRUE)
```

```
## lavaan 0.6-19 ended normally after 24 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        20
##
##   Number of observations                           175
##
## Model Test User Model:
##
##   Test statistic                                93.217
##   Degrees of freedom                                35
##   P-value (Chi-square)                           0.000
##
## Model Test Baseline Model:
##
##   Test statistic                               503.222
##   Degrees of freedom                                45
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.873
##   Tucker-Lewis Index (TLI)                       0.837
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -4078.655
##   Loglikelihood unrestricted model (H1)      -4032.046
##
##   Akaike (AIC)                                8197.309
##   Bayesian (BIC)                              8260.605
##   Sample-size adjusted Bayesian (SABIC)       8197.271
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.097
##   90 Percent confidence interval - lower         0.074
##   90 Percent confidence interval - upper         0.122
##   P-value H_0: RMSEA <= 0.050                    0.001
##   P-value H_0: RMSEA >= 0.080                    0.892
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.160
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
```

```
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   verbal =~
##     info              2.269    0.199   11.389    0.000    2.269    0.781
##     comp              1.958    0.214    9.160    0.000    1.958    0.662
##     arith             1.309    0.172    7.591    0.000    1.309    0.569
##     simil             2.179    0.227    9.589    0.000    2.179    0.686
##     vocab             2.277    0.201   11.335    0.000    2.277    0.779
##     digit             1.085    0.212    5.118    0.000    1.085    0.402
##   performance =~
##     pictcomp          1.664    0.255    6.518    0.000    1.664    0.569
##     parang            1.244    0.233    5.347    0.000    1.244    0.469
##     block             1.868    0.241    7.753    0.000    1.868    0.691
##     object            1.669    0.248    6.732    0.000    1.669    0.589
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .info              3.284    0.501    6.550    0.000    3.284    0.389
##    .comp              4.907    0.615    7.979    0.000    4.907    0.561
##    .arith             3.579    0.420    8.521    0.000    3.579    0.676
##    .simil             5.331    0.685    7.782    0.000    5.331    0.529
##    .vocab             3.366    0.510    6.599    0.000    3.366    0.394
##    .digit             6.093    0.675    9.021    0.000    6.093    0.838
##    .pictcomp          5.790    0.807    7.174    0.000    5.790    0.676
##    .parang            5.486    0.676    8.110    0.000    5.486    0.780
##    .block             3.813    0.730    5.224    0.000    3.813    0.522
##    .object            5.255    0.760    6.915    0.000    5.255    0.653
##     verbal            1.000                               1.000    1.000
##     performance       1.000                               1.000    1.000
```

```
lavaanPlot(fit2,
           coefs = TRUE,
           stand = TRUE)
```



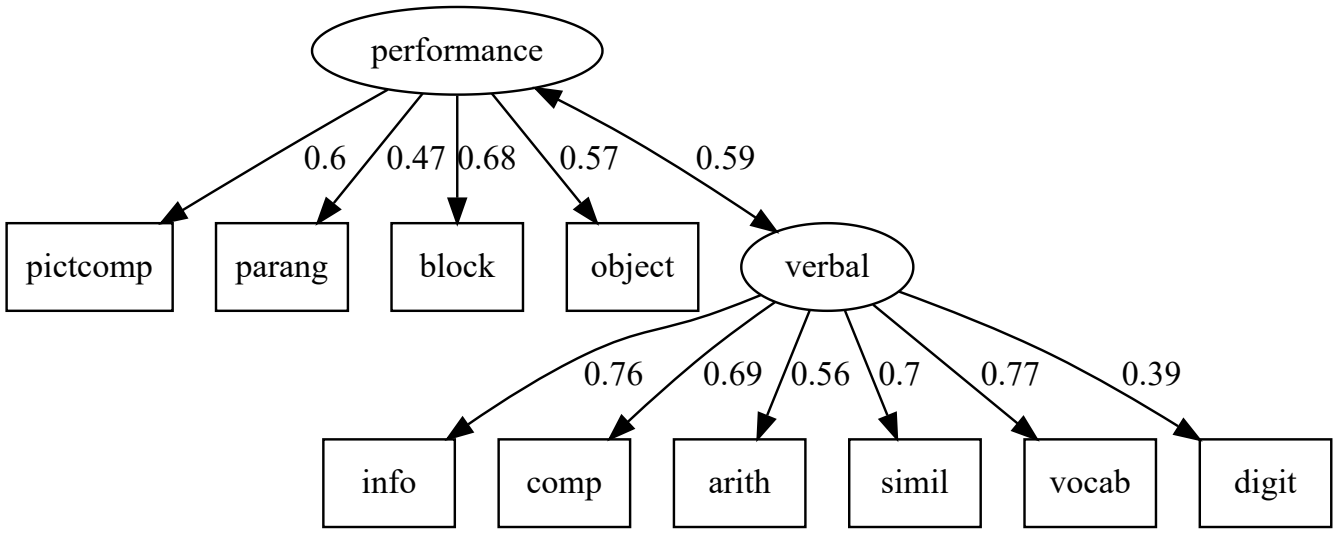

```
# Run the model: Two Factor Model
fit2c <- lavaan(ctwofactor,
                data = wiscsem,
                std.lv = TRUE)

# Get parameter estiamtes and fit statistics
summary(fit2c,
        fit.measures = TRUE,
        standardized = TRUE)
```

```
## lavaan 0.6-19 ended normally after 23 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        21
##
##   Number of observations                           175
##
## Model Test User Model:
##
##   Test statistic                                55.305
##   Degrees of freedom                                34
##   P-value (Chi-square)                           0.012
##
## Model Test Baseline Model:
##
##   Test statistic                               503.222
##   Degrees of freedom                                45
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.954
##   Tucker-Lewis Index (TLI)                       0.938
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -4059.699
##   Loglikelihood unrestricted model (H1)      -4032.046
##
##   Akaike (AIC)                                8161.397
##   Bayesian (BIC)                              8227.858
##   Sample-size adjusted Bayesian (SABIC)       8161.357
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.060
##   90 Percent confidence interval - lower         0.028
##   90 Percent confidence interval - upper         0.088
##   P-value H_0: RMSEA <= 0.050                    0.268
##   P-value H_0: RMSEA >= 0.080                    0.124
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.055
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
```

```
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   verbal =~
##     info              2.207    0.200   11.031    0.000    2.207    0.760
##     comp              2.042    0.210    9.710    0.000    2.042    0.691
##     arith             1.299    0.172    7.550    0.000    1.299    0.565
##     simil             2.234    0.224    9.951    0.000    2.234    0.704
##     vocab             2.249    0.201   11.218    0.000    2.249    0.769
##     digit             1.051    0.212    4.960    0.000    1.051    0.390
##   performance =~
##     pictcomp          1.761    0.242    7.266    0.000    1.761    0.602
##     parang            1.251    0.225    5.570    0.000    1.251    0.472
##     block             1.833    0.222    8.242    0.000    1.833    0.678
##     object            1.604    0.236    6.794    0.000    1.604    0.566
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   verbal ~~
##     performance       0.589    0.075    7.806    0.000    0.589    0.589
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .info              3.564    0.507    7.032    0.000    3.564    0.423
##    .comp              4.572    0.585    7.815    0.000    4.572    0.523
##    .arith             3.604    0.420    8.573    0.000    3.604    0.681
##    .simil             5.087    0.661    7.696    0.000    5.087    0.505
##    .vocab             3.491    0.507    6.891    0.000    3.491    0.408
##    .digit             6.165    0.681    9.057    0.000    6.165    0.848
##    .pictcomp          5.460    0.757    7.217    0.000    5.460    0.638
##    .parang            5.467    0.659    8.298    0.000    5.467    0.777
##    .block             3.943    0.641    6.155    0.000    3.943    0.540
##    .object            5.468    0.720    7.595    0.000    5.468    0.680
##     verbal            1.000                               1.000    1.000
##     performance       1.000                               1.000    1.000
```

```
lavaanPlot(fit2c,
           coefs = TRUE,
           stand = TRUE,
           covs = TRUE)
```



Direct Model Fit Comparisons (Chi-square Difference Test)

```
# Can compare fit of our 2-factor models because they are nested (same items) AND one has fewer degrees of freedom
anova(fit2,fit2c)
```

```
##
## Chi-Squared Difference Test
##
##       Df    AIC    BIC  Chisq Chisq diff   RMSEA Df diff Pr(>Chisq)
## fit2c 34 8161.4 8227.9 55.305
## fit2  35 8197.3 8260.6 93.217     37.912 0.45927       1  7.402e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(fit, fit2c)
```

```
##
## Chi-Squared Difference Test
##
##       Df    AIC    BIC   Chisq Chisq diff   RMSEA Df diff Pr(>Chisq)
## fit2c 34 8161.4 8227.9  55.305
## fit   35 8205.7 8269.0 101.635      46.33 0.50895       1  9.993e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Then we can call for the summary of the "best" model and the model fit indices
summary(fit2c,
        fit.measures = TRUE,
        standardized = TRUE)
```

```
## lavaan 0.6-19 ended normally after 23 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        21
##
##   Number of observations                           175
##
## Model Test User Model:
##
##   Test statistic                                55.305
##   Degrees of freedom                                34
##   P-value (Chi-square)                           0.012
##
## Model Test Baseline Model:
##
##   Test statistic                               503.222
##   Degrees of freedom                                45
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.954
##   Tucker-Lewis Index (TLI)                       0.938
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -4059.699
##   Loglikelihood unrestricted model (H1)      -4032.046
##
##   Akaike (AIC)                                8161.397
##   Bayesian (BIC)                              8227.858
##   Sample-size adjusted Bayesian (SABIC)       8161.357
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.060
##   90 Percent confidence interval - lower         0.028
##   90 Percent confidence interval - upper         0.088
##   P-value H_0: RMSEA <= 0.050                    0.268
##   P-value H_0: RMSEA >= 0.080                    0.124
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.055
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
```

```
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   verbal =~
##     info              2.207    0.200   11.031    0.000    2.207    0.760
##     comp              2.042    0.210    9.710    0.000    2.042    0.691
##     arith             1.299    0.172    7.550    0.000    1.299    0.565
##     simil             2.234    0.224    9.951    0.000    2.234    0.704
##     vocab             2.249    0.201   11.218    0.000    2.249    0.769
##     digit             1.051    0.212    4.960    0.000    1.051    0.390
##   performance =~
##     pictcomp          1.761    0.242    7.266    0.000    1.761    0.602
##     parang            1.251    0.225    5.570    0.000    1.251    0.472
##     block             1.833    0.222    8.242    0.000    1.833    0.678
##     object            1.604    0.236    6.794    0.000    1.604    0.566
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   verbal ~~
##     performance       0.589    0.075    7.806    0.000    0.589    0.589
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .info              3.564    0.507    7.032    0.000    3.564    0.423
##    .comp              4.572    0.585    7.815    0.000    4.572    0.523
##    .arith             3.604    0.420    8.573    0.000    3.604    0.681
##    .simil             5.087    0.661    7.696    0.000    5.087    0.505
##    .vocab             3.491    0.507    6.891    0.000    3.491    0.408
##    .digit             6.165    0.681    9.057    0.000    6.165    0.848
##    .pictcomp          5.460    0.757    7.217    0.000    5.460    0.638
##    .parang            5.467    0.659    8.298    0.000    5.467    0.777
##    .block             3.943    0.641    6.155    0.000    3.943    0.540
##    .object            5.468    0.720    7.595    0.000    5.468    0.680
##     verbal            1.000                               1.000    1.000
##     performance       1.000                               1.000    1.000
```

```
fitMeasures(fit2c)
```

```
##                  npar                  fmin                 chisq
##                21.000                 0.158                55.305
##                    df                pvalue        baseline.chisq
##                34.000                 0.012               503.222
##           baseline.df       baseline.pvalue                   cfi
##                45.000                 0.000                 0.954
##                   tli                  nnfi                   rfi
##                 0.938                 0.938                 0.855
##                   nfi                  pnfi                   ifi
##                 0.890                 0.673                 0.955
##                   rni                  logl     unrestricted.logl
##                 0.954             -4059.699             -4032.046
##                   aic                   bic                ntotal
##              8161.397              8227.858               175.000
##                  bic2                 rmsea        rmsea.ci.lower
##              8161.357                 0.060                 0.028
##        rmsea.ci.upper        rmsea.ci.level          rmsea.pvalue
##                 0.088                 0.900                 0.268
##        rmsea.close.h0 rmsea.notclose.pvalue     rmsea.notclose.h0
##                 0.050                 0.124                 0.080
##                   rmr            rmr_nomean                  srmr
##                 0.439                 0.439                 0.055
##          srmr_bentler   srmr_bentler_nomean                  crmr
##                 0.055                 0.055                 0.061
##           crmr_nomean            srmr_mplus     srmr_mplus_nomean
##                 0.061                 0.055                 0.055
##                 cn_05                 cn_01                   gfi
##               154.790               178.391                 0.940
##                  agfi                  pgfi                   mfi
##                 0.902                 0.581                 0.941
##                  ecvi
##                 0.556
```

Structural Equation Model

Model Specification (Step 1 SEM: Measurement Model)

Correlated Three Factor Model:

```
cthreefactor <- ' # Latent variables
Fluid =~ V1 + V2 + V3
Verbal =~ V4 + V5 + V6
Visuospatial =~ V7 + V8 + V9

# Latent variable correlation
Fluid ~~ Verbal
Fluid ~~ Visuospatial
Verbal ~~ Visuospatial

# Residuals for manifest variables
V1 ~~ V1
V2 ~~ V2
V3 ~~ V3
V4 ~~ V4
V5 ~~ V5
V6 ~~ V6
V7 ~~ V7
V8 ~~ V8
V9 ~~ V9

# Variances for latent variables
Fluid~~1*Fluid
Verbal~~1*Verbal
Visuospatial~~1*Visuospatial
'
```

Model Specification (Step 2 SEM: Full Structural Model)

Fluid Intelligence Predicts Verbal/Visuospatial Model:

```
gfpredicts <- ' # Latent variables
Fluid =~ V1 + V2 + V3
Verbal =~ V4 + V5 + V6
Visuospatial =~ V7 + V8 + V9

# Regressions

Verbal + Visuospatial ~ Fluid

# Residuals for manifest variables
V1 ~~ V1
V2 ~~ V2
V3 ~~ V3
V4 ~~ V4
V5 ~~ V5
V6 ~~ V6
V7 ~~ V7
V8 ~~ V8
V9 ~~ V9

# Variances for latent variables
Fluid~~1*Fluid
Verbal~~1*Verbal
Visuospatial~~1*Visuospatial
'
```

Fluid Intelligence Predicts Verbal/Visuospatial Correlated Model:

```
gfpredictsc <- ' # Latent variables
Fluid =~ V1 + V2 + V3
Verbal =~ V4 + V5 + V6
Visuospatial =~ V7 + V8 + V9

# Correlated Latent variables
Verbal ~~ Visuospatial

# Regressions

Verbal + Visuospatial ~ Fluid

# Residuals for manifest variables
V1 ~~ V1
V2 ~~ V2
V3 ~~ V3
V4 ~~ V4
V5 ~~ V5
V6 ~~ V6
V7 ~~ V7
V8 ~~ V8
V9 ~~ V9

# Variances for latent variables
Fluid~~1*Fluid
Verbal~~1*Verbal
Visuospatial~~1*Visuospatial
'
```

Run the CFA measurement model, summarize the parameter estimates, and plot

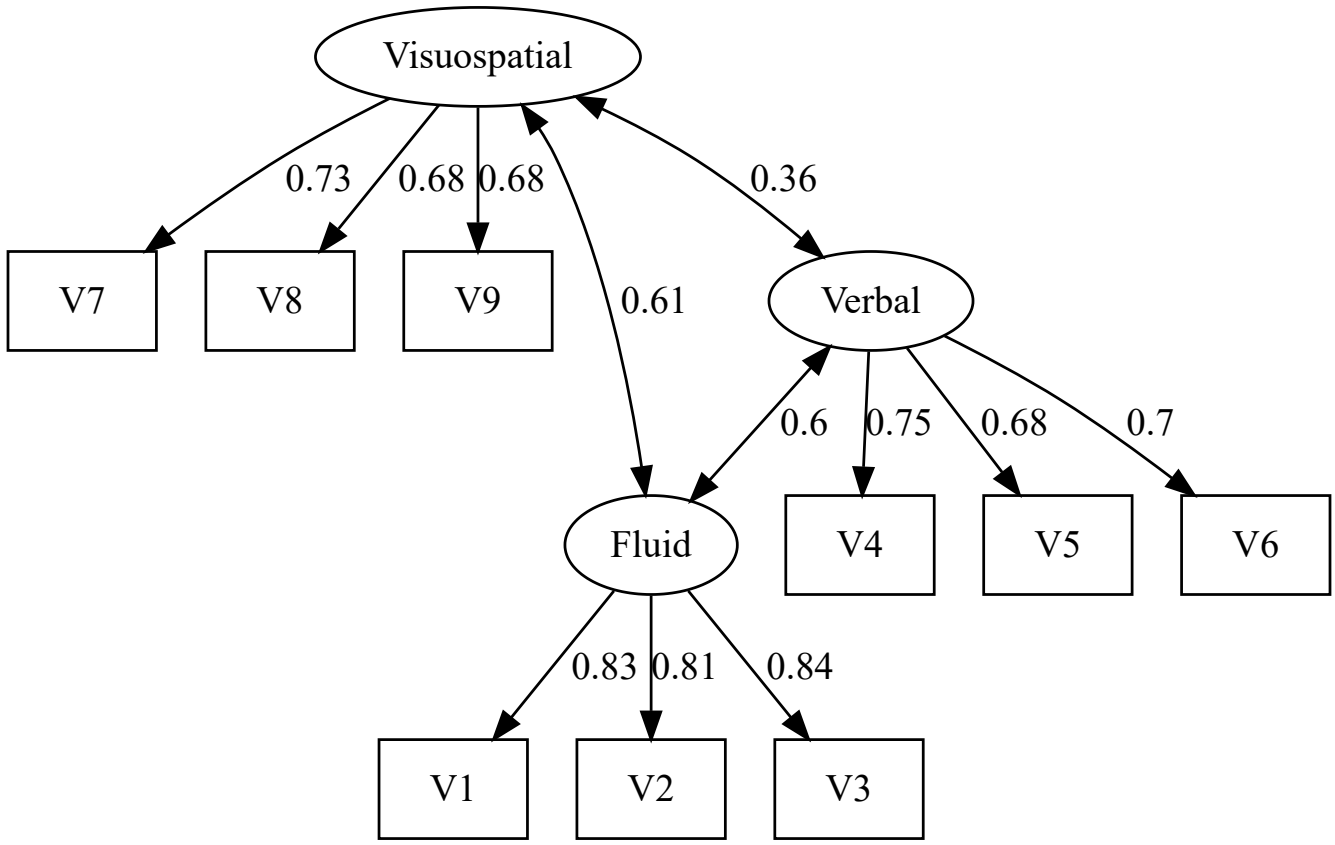
```
# Run the model: Three Factor Model
cfafit <- lavaan(cthreefactor,
                 data = cogability,
                 std.lv = TRUE)

# Get parameter estiamtes and fit statistics
summary(cfafit,
        fit.measures = TRUE,
        standardized = TRUE)
```

```
## lavaan 0.6-19 ended normally after 23 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        21
##
##   Number of observations                           250
##
## Model Test User Model:
##
##   Test statistic                                31.927
##   Degrees of freedom                                24
##   P-value (Chi-square)                           0.129
##
## Model Test Baseline Model:
##
##   Test statistic                               876.448
##   Degrees of freedom                                36
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.991
##   Tucker-Lewis Index (TLI)                       0.986
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -9523.464
##   Loglikelihood unrestricted model (H1)      -9507.500
##
##   Akaike (AIC)                               19088.927
##   Bayesian (BIC)                             19162.878
##   Sample-size adjusted Bayesian (SABIC)      19096.306
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.036
##   90 Percent confidence interval - lower         0.000
##   90 Percent confidence interval - upper         0.067
##   P-value H_0: RMSEA <= 0.050                    0.738
##   P-value H_0: RMSEA >= 0.080                    0.006
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.032
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
```

```
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Fluid =~
##     V1               15.442    1.007   15.333    0.000   15.442    0.832
##     V2               16.124    1.088   14.824    0.000   16.124    0.812
##     V3               16.311    1.041   15.664    0.000   16.311    0.844
##   Verbal =~
##     V4               16.213    1.358   11.943    0.000   16.213    0.754
##     V5               14.416    1.351   10.669    0.000   14.416    0.682
##     V6               13.854    1.268   10.924    0.000   13.854    0.696
##   Visuospatial =~
##     V7               15.075    1.317   11.450    0.000   15.075    0.734
##     V8               14.152    1.342   10.546    0.000   14.152    0.681
##     V9               13.294    1.266   10.500    0.000   13.294    0.679
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Fluid ~~
##     Verbal            0.597    0.057   10.563    0.000    0.597    0.597
##     Visuospatial      0.610    0.057   10.795    0.000    0.610    0.610
##   Verbal ~~
##     Visuospatial      0.358    0.077    4.656    0.000    0.358    0.358
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .V1              106.192   14.060    7.553    0.000  106.192    0.308
##    .V2              133.971   16.609    8.066    0.000  133.971    0.340
##    .V3              107.237   14.940    7.178    0.000  107.237    0.287
##    .V4              199.869   29.393    6.800    0.000  199.869    0.432
##    .V5              239.473   28.891    8.289    0.000  239.473    0.535
##    .V6              204.332   25.431    8.035    0.000  204.332    0.516
##    .V7              194.623   27.530    7.070    0.000  194.623    0.461
##    .V8              231.052   28.431    8.127    0.000  231.052    0.536
##    .V9              206.917   25.315    8.174    0.000  206.917    0.539
##     Fluid             1.000                               1.000    1.000
##     Verbal            1.000                               1.000    1.000
##     Visuospatial      1.000                               1.000    1.000
```

```
lavaanPlot(cfafit,
           coefs = TRUE,
           stand = TRUE,
           covs = TRUE)
```



Run the SEM model(s), summarize the parameter estimates, and plot

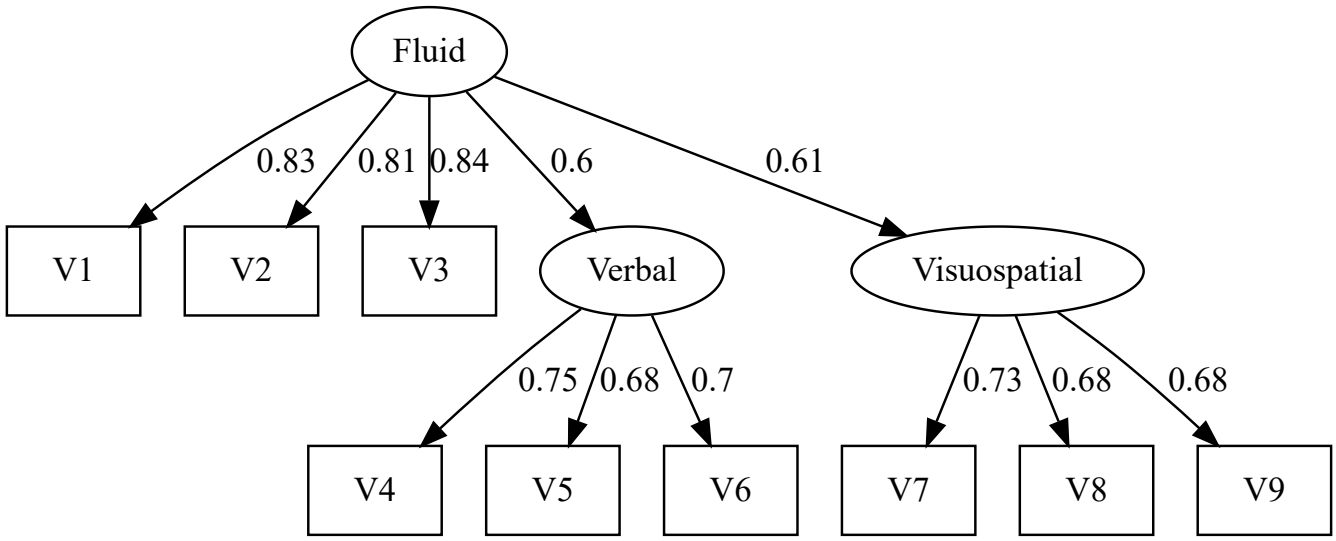
```
# Run the model: Gf Predicts SEM
semfit <- lavaan(gfpredicts,
                 data = cogability,
                 std.lv = TRUE)

# Get parameter estiamtes and fit statistics
summary(semfit,
        fit.measures = TRUE,
        standardized = TRUE)
```

```
## lavaan 0.6-19 ended normally after 21 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        20
##
##   Number of observations                           250
##
## Model Test User Model:
##
##   Test statistic                                31.937
##   Degrees of freedom                                25
##   P-value (Chi-square)                           0.160
##
## Model Test Baseline Model:
##
##   Test statistic                               876.448
##   Degrees of freedom                                36
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.992
##   Tucker-Lewis Index (TLI)                       0.988
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -9523.469
##   Loglikelihood unrestricted model (H1)      -9507.500
##
##   Akaike (AIC)                               19086.938
##   Bayesian (BIC)                             19157.367
##   Sample-size adjusted Bayesian (SABIC)      19093.965
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.033
##   90 Percent confidence interval - lower         0.000
##   90 Percent confidence interval - upper         0.064
##   P-value H_0: RMSEA <= 0.050                    0.787
##   P-value H_0: RMSEA >= 0.080                    0.004
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.032
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
```

```
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Fluid =~
##     V1               15.440    1.007   15.329    0.000   15.440    0.832
##     V2               16.127    1.088   14.828    0.000   16.127    0.813
##     V3               16.314    1.041   15.669    0.000   16.314    0.844
##   Verbal =~
##     V4               13.006    1.220   10.660    0.000   16.206    0.753
##     V5               11.572    1.176    9.841    0.000   14.420    0.682
##     V6               11.120    1.109   10.028    0.000   13.857    0.696
##   Visuospatial =~
##     V7               11.960    1.176   10.169    0.000   15.085    0.734
##     V8               11.213    1.168    9.602    0.000   14.143    0.681
##     V9               10.539    1.101    9.573    0.000   13.293    0.679
##
## Regressions:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Verbal ~
##     Fluid             0.743    0.109    6.828    0.000    0.597    0.597
##   Visuospatial ~
##     Fluid             0.769    0.113    6.810    0.000    0.609    0.609
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .V1              106.272   14.063    7.557    0.000  106.272    0.308
##    .V2              133.869   16.606    8.061    0.000  133.869    0.340
##    .V3              107.113   14.934    7.172    0.000  107.113    0.287
##    .V4              200.086   29.394    6.807    0.000  200.086    0.432
##    .V5              239.352   28.889    8.285    0.000  239.352    0.535
##    .V6              204.260   25.431    8.032    0.000  204.260    0.515
##    .V7              194.341   27.531    7.059    0.000  194.341    0.461
##    .V8              231.300   28.435    8.134    0.000  231.300    0.536
##    .V9              206.961   25.316    8.175    0.000  206.961    0.539
##     Fluid             1.000                               1.000    1.000
##    .Verbal            1.000                               0.644    0.644
##    .Visuospatial      1.000                               0.629    0.629
```

```
lavaanPlot(semfit,
           coefs = TRUE,
           stand = TRUE,
           covs = TRUE)
```



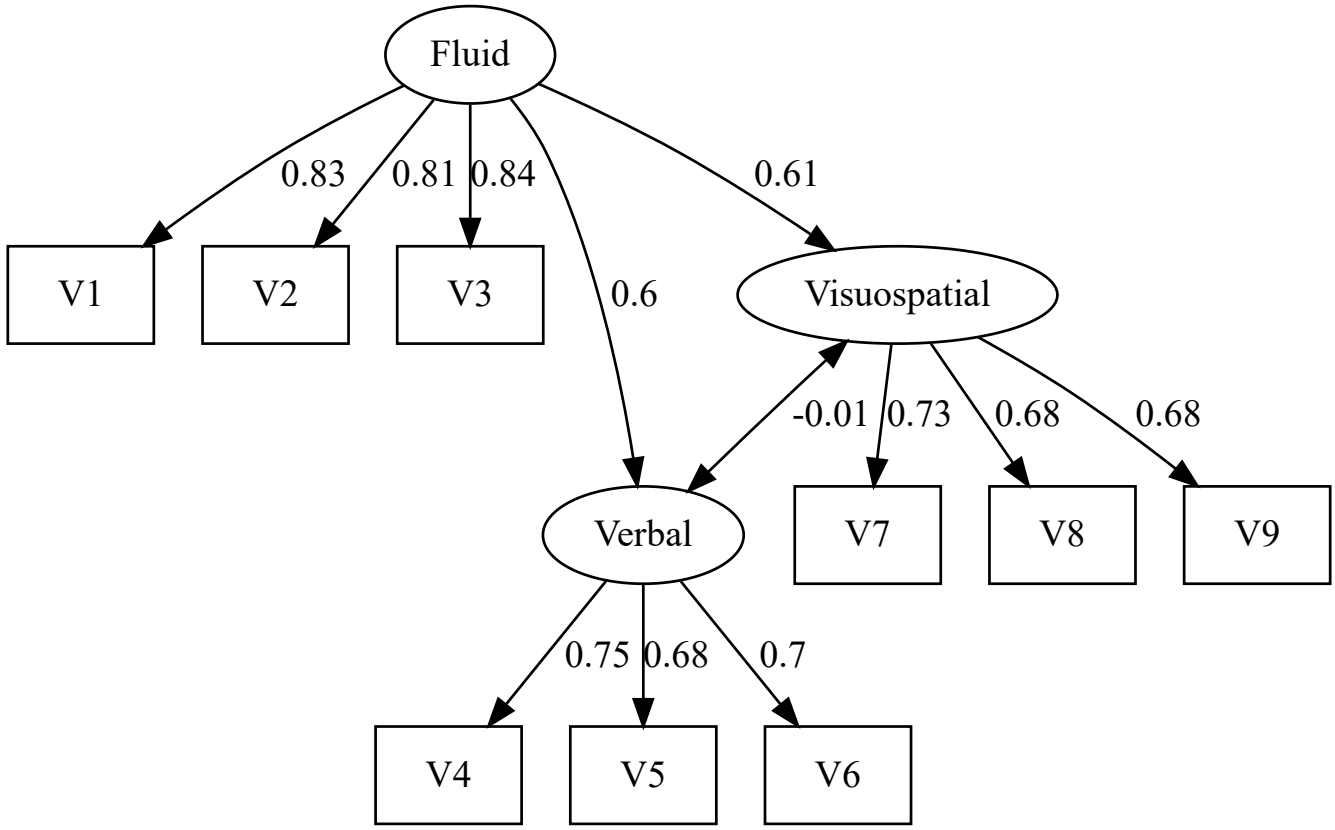

```
# Run the model: Gf Predicts Correlated SEM
semfitc <- lavaan(gfpredictsc,
                  data = cogability,
                  std.lv = TRUE)

# Get parameter estiamtes and fit statistics
summary(semfitc,
        fit.measures = TRUE,
        standardized = TRUE)
```

```
## lavaan 0.6-19 ended normally after 21 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        21
##
##   Number of observations                           250
##
## Model Test User Model:
##
##   Test statistic                                31.927
##   Degrees of freedom                                24
##   P-value (Chi-square)                           0.129
##
## Model Test Baseline Model:
##
##   Test statistic                               876.448
##   Degrees of freedom                                36
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.991
##   Tucker-Lewis Index (TLI)                       0.986
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -9523.464
##   Loglikelihood unrestricted model (H1)      -9507.500
##
##   Akaike (AIC)                               19088.927
##   Bayesian (BIC)                             19162.878
##   Sample-size adjusted Bayesian (SABIC)      19096.306
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.036
##   90 Percent confidence interval - lower         0.000
##   90 Percent confidence interval - upper         0.067
##   P-value H_0: RMSEA <= 0.050                    0.738
##   P-value H_0: RMSEA >= 0.080                    0.006
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.032
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
```

```
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Fluid =~
##     V1               15.442    1.007   15.333    0.000   15.442    0.832
##     V2               16.124    1.088   14.824    0.000   16.124    0.812
##     V3               16.311    1.041   15.664    0.000   16.311    0.844
##   Verbal =~
##     V4               13.004    1.222   10.642    0.000   16.213    0.754
##     V5               11.562    1.177    9.822    0.000   14.416    0.682
##     V6               11.112    1.110   10.009    0.000   13.854    0.696
##   Visuospatial =~
##     V7               11.945    1.178   10.144    0.000   15.075    0.734
##     V8               11.213    1.169    9.590    0.000   14.152    0.681
##     V9               10.534    1.102    9.557    0.000   13.294    0.679
##
## Regressions:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Verbal ~
##     Fluid             0.745    0.110    6.795    0.000    0.597    0.597
##   Visuospatial ~
##     Fluid             0.770    0.114    6.777    0.000    0.610    0.610
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##  .Verbal ~~
##    .Visuospatial     -0.011    0.102   -0.104    0.918   -0.011   -0.011
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .V1              106.192   14.060    7.553    0.000  106.192    0.308
##    .V2              133.971   16.609    8.066    0.000  133.971    0.340
##    .V3              107.237   14.940    7.178    0.000  107.237    0.287
##    .V4              199.868   29.393    6.800    0.000  199.868    0.432
##    .V5              239.473   28.891    8.289    0.000  239.473    0.535
##    .V6              204.331   25.431    8.035    0.000  204.331    0.516
##    .V7              194.623   27.530    7.070    0.000  194.623    0.461
##    .V8              231.052   28.431    8.127    0.000  231.052    0.536
##    .V9              206.917   25.315    8.174    0.000  206.917    0.539
##     Fluid             1.000                               1.000    1.000
##    .Verbal            1.000                               0.643    0.643
##    .Visuospatial      1.000                               0.628    0.628
```

```
lavaanPlot(semfitc,
           coefs = TRUE,
           stand = TRUE,
           covs = TRUE)
```



Direct Model Fit Comparisons (Chi-square Difference Test)

```
# Can compare fit of our 2-factor models because they are nested (same items) AND one has fewer degrees of freedom
anova(semfit,semfitc)
```

```
##
## Chi-Squared Difference Test
##
##         Df   AIC   BIC  Chisq Chisq diff RMSEA Df diff Pr(>Chisq)
## semfitc 24 19089 19163 31.927
## semfit  25 19087 19157 31.937   0.010397     0       1     0.9188
```